

triacylglycerol with oleic acid at the *sn*-2 position (Table 2). When triacylglycerol containing oleic acid at the *sn*-2 position is acidolysed with oleic acid using a specific lipase of *sn*-1,3 will produce triolein (Wang *et al.*, 2020), which is not a *sn*-2 palmitate product. Table 3 shows that palm stearin has palmitic acid content at the *sn*-2 position ranging from 23.0%-70.1%. Thus, the palmitic acid content at the *sn*-2 position in palm stearin needs to be increased to produce a good substrate for HMFS synthesis (Hasibuan *et al.*, 2021b).

Technologies for improving the positioning of palmitic acid at the *sn*-2 position of palm stearin are solvent fractionation (Ghosh *et al.*, 2016; Lee *et al.*, 2010; Wang *et al.*, 2019; Zou *et al.*, 2012a), enzymatic interesterification (Jiménez *et al.*, 2010a; 2010b) or chemical interesterification (Zou *et al.*, 2011; Zou

et al., 2012b). Palm stearin is fractionated using acetone as a solvent to produce a tripalmitin-rich triacylglycerol (92.0%) (Lee *et al.*, 2010) and palmitic acid (88.57%-92.30%) (Ghosh *et al.*, 2016; Wang *et al.*, 2019; Zou *et al.*, 2012a). In general, the palm stearin solvent fractionation process condition is carried out at an acetone ratio of 5-9 and a fractionation temperature of 20°C-40°C for 3-24 hr.

Enzymatic interesterification between palm stearin (60.0% palmitic acid and 23.0% palmitic acid at the *sn*-2 position) with palmitic acid using Novozyme 435 produces a product with 68.0%-75.0% palmitic acid at the *sn*-2 position (Jiménez *et al.*, 2010a; 2010b). Meanwhile, chemical interesterification of palm stearin (41.7% palmitic acid at the *sn*-2 position) resulted in a product with 58.0% palmitic acid at the *sn*-2 position (Zou *et al.*,

TABLE 2. TRIACYLGLYCEROLS COMPOSITION OF PALM OIL FRACTIONS

Triacylglycerols (%)	Palm oil*	Palm olein*	Palm stearin*	Soft palm stearin**	Hard palm stearin**	Palm kernel oil*
CCLa						6.8
CLaLa						9.9
LaLaLa						21.2
LaLaM						17.0
LaLaO						5.3
LaMM						8.8
PLL				1.0	ND	
MMM	0.4	0.6	0.2	ND	ND	
LaLaP						1.2
LaMO						4.6
MPL	2.4	3.7	1.0	0.1	ND	
LaMP						4.6
LaOO						3.8
LaPO						4.3
LaPP+MMO						
OOL	0.7	0.8	0.1	0.8	ND	
MMP	1.8	2.6	0.8	ND	ND	0.7
MOO						2.0
POL	10.1	15.8	5.3	6.9	0.9	
PPL	9.8	11.2	7.8	8.1	3.8	0.6
MPP	0.6	ND	2.3	ND	ND	
OOO	4.1	5.6	1.8	4.1	4.5	1.4
POO	24.2	36.3	12.0	18.0	3.0	1.9
PPO	31.1	17.1	29.8	32.0	26.4	1.1
PPP	5.9	0.1	29.2	20.6	51.0	0.1
SOO	2.3	3.6	0.8	ND	ND	0.4
PSO	5.1	2.5	3.8	4.9	2.9	0.4
PPS	0.9	ND	5.2	3.5	7.4	
SSO	0.5	ND	ND	ND	ND	

Note: ND - not detected; L - lauric acid; M-myristic acid; O - oleic acid; P - palmitic acid; S - stearic acid.

Source: *Tan and Man (2002); ** Ibrahim *et al.* (2006).